THE CHINESE ECONOMY

A JOURNEY INTO HISTORY, ECONOMICS, AND POLICY

Lecture 3: How do economists estimate causal effects? From Mao to the reforms

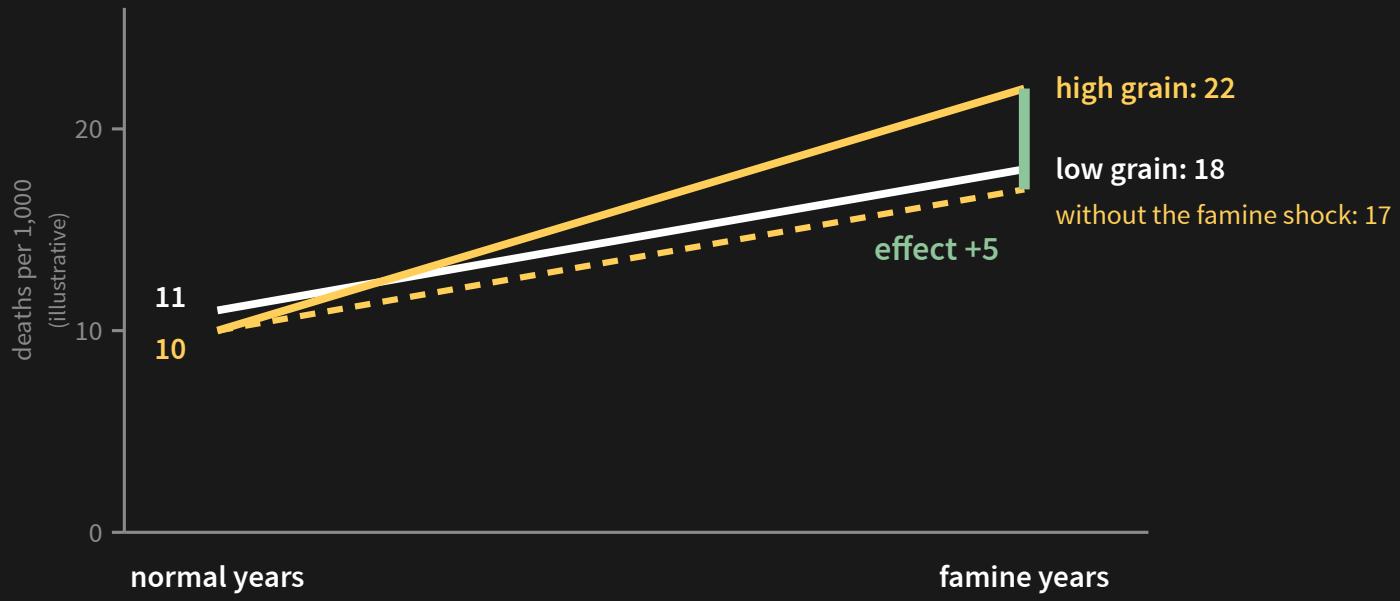
LAST WEEK, BY HAND

On the board: the table of last week. Deaths per 1,000 people, illustrative numbers.

	Normal years	Famine years	Change
High expected grain	10	22	+12
Low expected grain	11	18	+7

Difference in differences: +12 − 7 = +5. But why may we subtract the low-grain change?

THE ASSUMPTION



Parallel trends: without the treatment, the two groups would have ***moved together***.
Their gap may exist, but — net of the treatment — it would stay **fixed over time**.

It is an **Assumption**: it has to be true if we want to retrieve a causal effect.

1961-76 — READJUSTMENT, THIRD FRONT, CULTURAL REVOLUTION

- **1961-63** readjustment: investment chopped back, about 20 million workers sent to the countryside
- **1964-66** Third Front: a new expansion into the interior
- **1966-76** Cultural Revolution: 17 million urban school-leavers sent to the countryside (*xiafang*); schools and universities closed
- **1976**: Mao dies

1968-78: YOUNG PEOPLE SENT TO THE COUNTRYSIDE

- About **16-17 million urban youths** were sent to live and work in villages during the Cultural Revolution
- Who went was not random: **family background**, schooling, politics, and household rules (often one child could stay in the city)
- Most came back after 1978

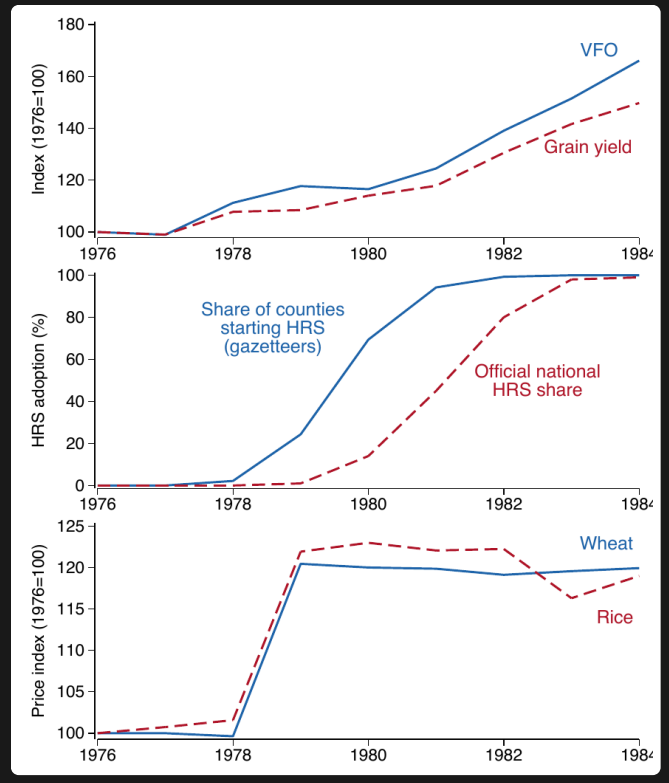
Question: did the years in the countryside change their lives?

1976-78 — THE TURN

- Growth had accelerated under the Big Push — but even more in Taiwan and Hong Kong, far ahead of the PRC by 1978
- That demonstration effect convinced Chinese leaders to reform and open up
- **December 1978**: Third Plenum — ease the terms of trade with agriculture, "give farmers a chance to catch their breath"
- Deng Xiaoping consolidates power; *gaige kaifang*, reform and opening

1978-84: THE LAND GOES BACK TO HOUSEHOLDS

- **Household Responsibility System**: the collective's land is contracted to families; above the quota, they keep what they grow
- First experiments: **Anhui, 1978**, after a severe drought. Then province by province, county by county, until 1984
- Meanwhile, **1979**: the state raises the price it pays for quota grain by about 20%
- Official grain yields: **+43%** from 1978 to 1984
- The reform did not arrive everywhere at once: a **staggered rollout**. For an economist, a gift



Ferguson and Kim (2026), Fig. 1: farm output and yields; share of counties that started the reform; procurement prices (1976 = 100)

WHAT WAS THE CAUSAL EFFECT OF THE TRANSITION?

- The ideal experiment: **two Chinas** in 1978, identical; one reforms, one does not
- There is **one** China. And everything changed **at once**: land, prices, leaders, trade, the opening to the world
- No comparison group, no lottery, no counterfactual we can approach

Whereof one cannot speak, thereof one must be silent.

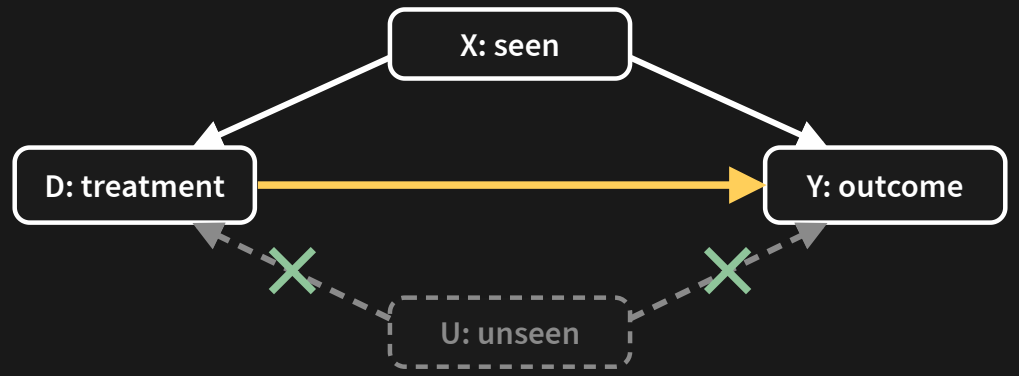
Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*, proposition 7 (1921; English translation by C. K. Ogden, 1922)

…unless we cut the question into pieces we can answer

PIECES WE CAN ANSWER

Question	Who could be compared with whom?	Method	Paper
Did the years in the countryside change the sent-down youths' lives?	brothers and sisters: one sent, one not	**matching**	Xie, Jiang, Greenman, "Did Send-down Experience Benefit Youth?" (2008)
Did giving the land back to households change who was born?	families in the same county and year, before and after the reform	**difference in differences**, year by year; nature's lottery	Almond, Li, Zhang, "Land Reform and Sex Selection in China" (2019)
Did it raise yields?	early and late reformers alike in what we see; fields just across a provincial border, before and after	**regression, propensity score**; difference in differences at the border	Ferguson, Kim, "What Started China's Miracle? The View from Outer Space" (2026)

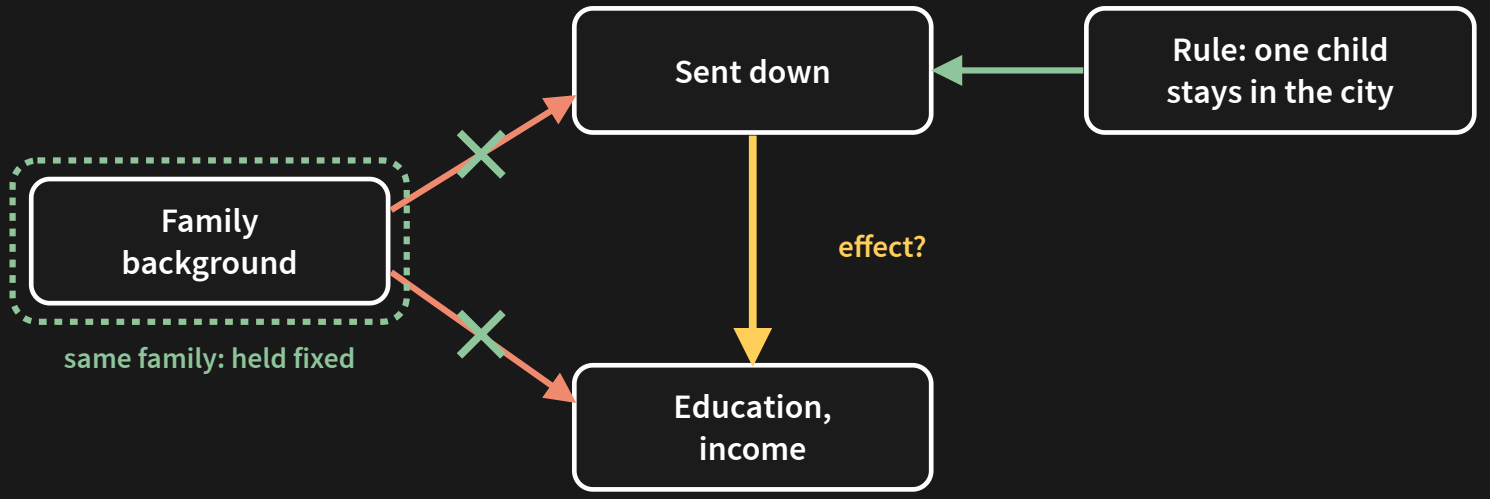
THREE WAYS TO CLOSE THE BACK DOOR



DiD: U cancels, if it does not change over time

Strategy	What it does in the graph	What we must assume
Lottery	cuts every arrow into D	the draw is really random
Matching	holds fixed the X we can see: compare like with like	no back door runs through what we cannot see
Difference in differences	lets an unseen U through, if it does not change over time	parallel trends

MATCHING: COMPARE LIKE WITH LIKE



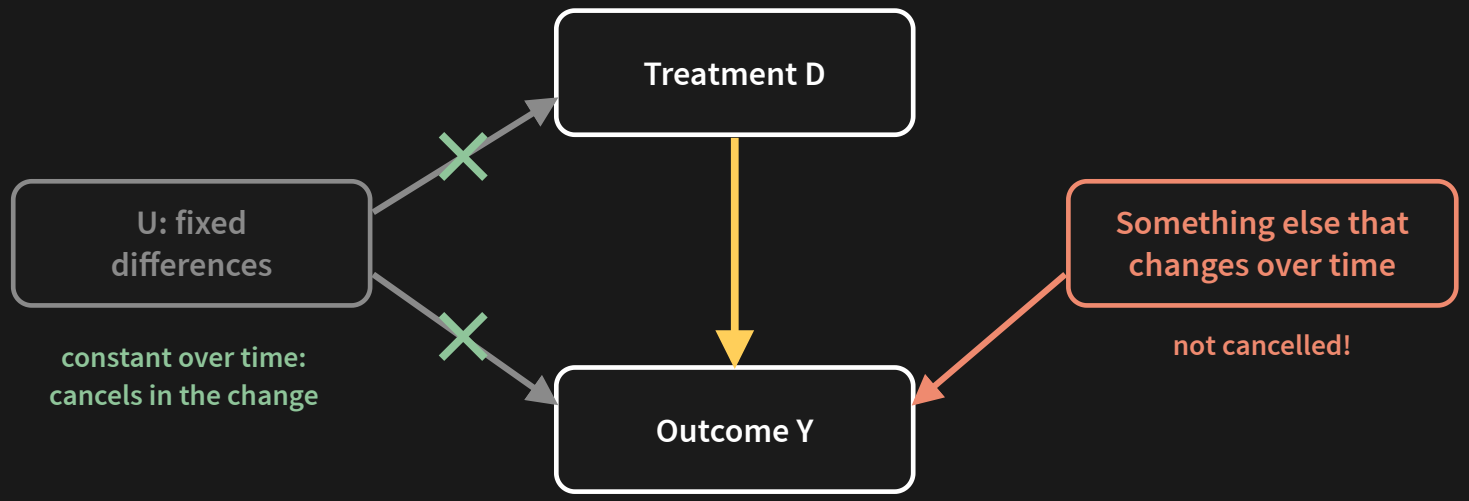
Compare a sent-down youth with someone **as similar as possible** who stayed: same city, same age... ideally **the same family**

SENT-DOWN YOUTHS: TWO COMPARISONS

- **Naive**: among people who were sent down for a short time, **15.2%** reached college; among those never sent down, **10.9%**. Was the countryside good for them?
- **Siblings**: 344 pairs of brothers and sisters, one sent down and one not. **No discernible difference** in any of four measures of socioeconomic status
- The first gap was **selection**, not an effect

Xie, Jiang and Greenman (2008), *Social Science Research* 37(2): survey of 1999 in Shanghai, Wuhan and Xi'an

DIFFERENCE IN DIFFERENCES, IN THE GRAPH



DiD survives what we cannot see, **if it stays put**. It fails with what **moves at the same time** as the treatment

LAND REFORM AND SEX SELECTION IN CHINA

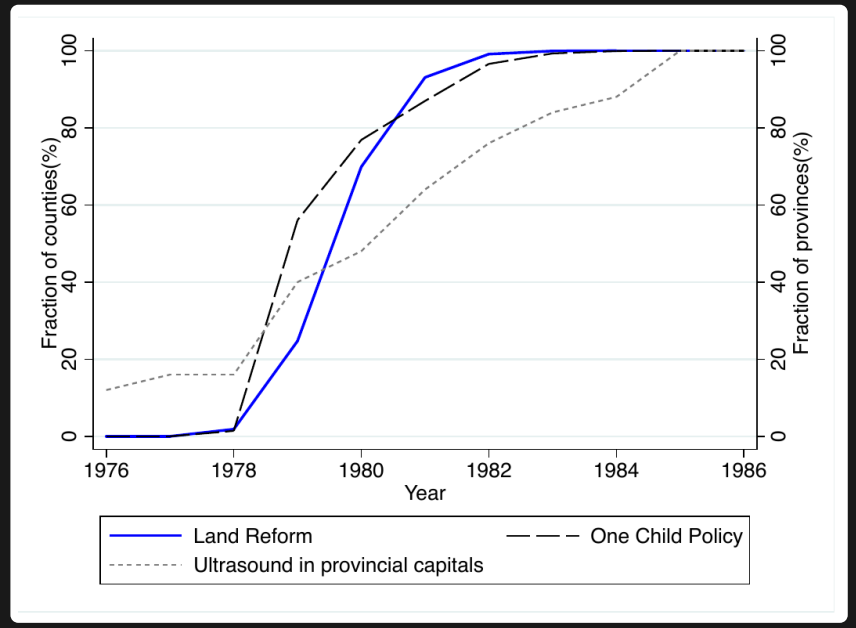
Douglas Almond, Hongbin Li, Shuang Zhang

JPE, 2019

- In a normal population about **105 boys** are born for every 100 girls
- In China the sex ratio at birth rose from **1.06 in 1978** to **1.20 in 2000**: hundreds of thousands of "missing girls" every year
- The usual suspects: ultrasound made sex selection cheaper; families had fewer children (the One Child Policy)
- But the rise **started around 1980**, when ultrasound was rare in the countryside and fertility was roughly flat. **Something else** started then
- The candidate: the **land reform** of 1978-84, which raised rural incomes

QUESTION AND DATA

- **Question**: did giving land back to households increase sex selection?
- **Treatment**: the year the reform started in each county, from the **county histories** (gazetteers) of 914 counties, half of rural China. The same histories give the year the One Child Policy started
- **Outcome**: the sex of children born 1974-86, from the **1990 census**. Within each family, children are ordered by birth date: first child, second child...
- Few people had moved (hukou): children are counted where they were born



Almond, Li and Zhang, Fig. 2: share of counties that had started land reform (solid) and the One Child Policy (dashed); provinces with ultrasound in the capital (dotted)

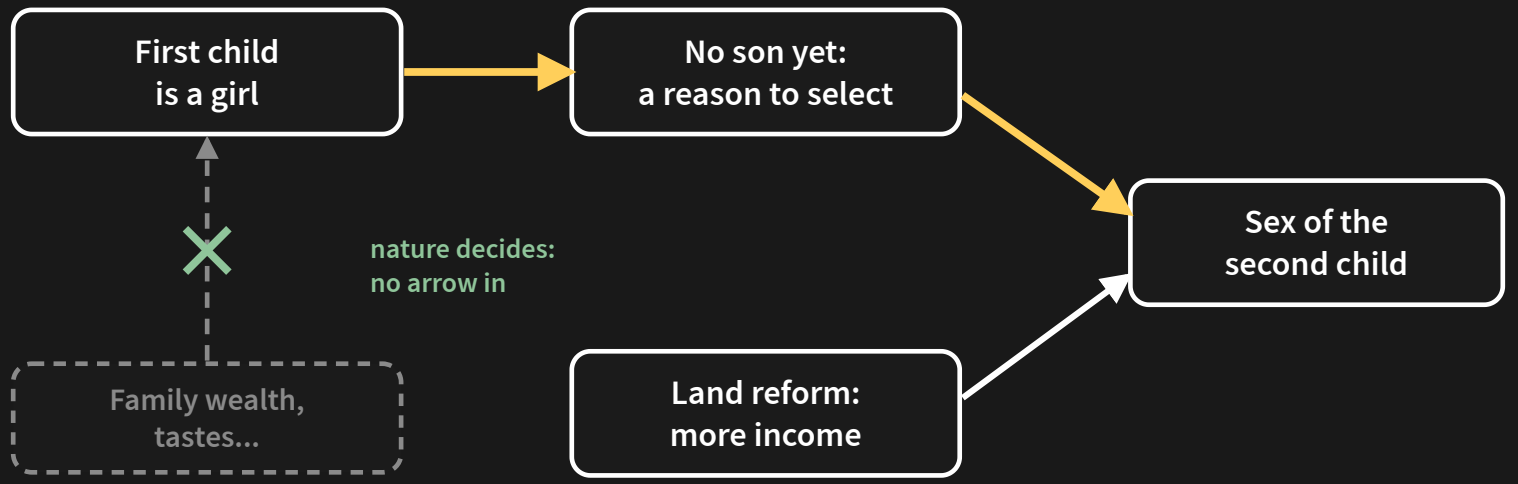
STRATEGY

Line up each county on **its own** reform year (year 0). Then compare, in the **same county and the same year**, three kinds of births:

Share of boys among...	Before reform	After reform	Change
first children (nature decides)	0.51	0.51	0
second children **after a boy** (little reason to select)	0.50	0.50	0
second children **after a girl** (a reason to select)	0.53	0.56	**+0.03**

- **Difference in differences**: the change after a girl (+3 points) minus the change after a boy (0). A shock that hit the whole county that year moves both rows
- The assumption: without the reform, girl-first and boy-first families would have kept the **same gap** (parallel trends)
- Year by year instead of before/after: an **event study**. Flat lines before year 0 make the assumption believable

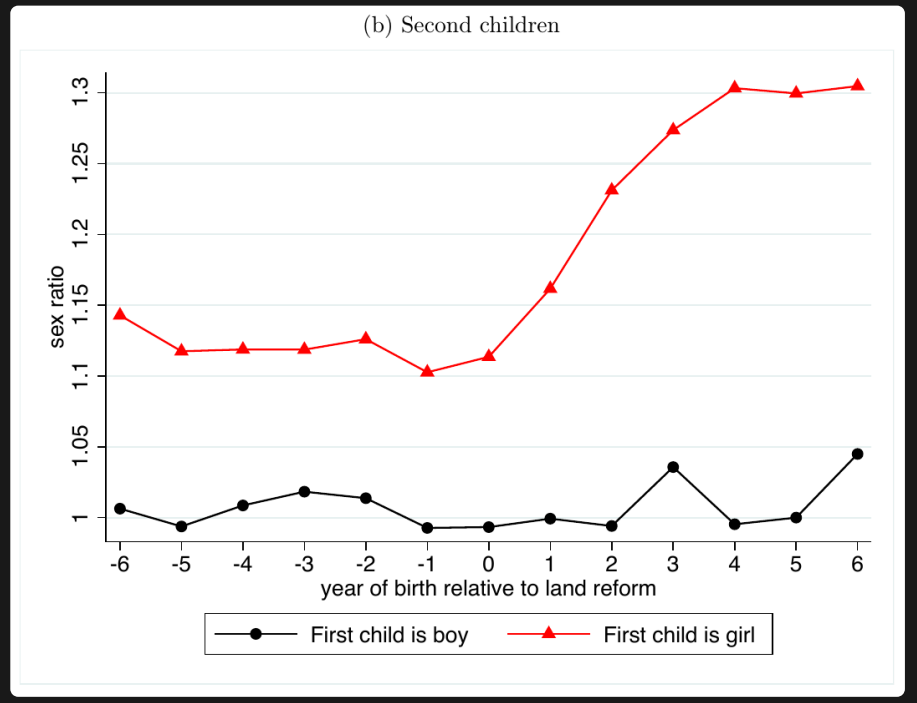
NATURE'S LOTTERY



- The sex of the first child is decided by **nature**: about 1.05 boys per girl, before and after the reform
- Nothing about the family pushes it one way or the other: no arrow comes in
- It moves the **reason to select**, and touches the second child only through it. That is the logic of an **instrument**
- The threat: families with a first girl being different in other ways. Normal first births (1.05) are the evidence against it

RESULTS

- Red: second children after a **first girl**. Flat before the reform, then from about **1.1 to 1.3 boys per girl** within four years
- Black: second children after a **first boy**. Nothing happens
- First children (not shown): stable at about 1.05
- Size: about **half** of the rise in rural sex ratios 1978-86, roughly **one million missing girls**



Almond, Li and Zhang, Fig. 4b: sex ratio of second children by year of birth relative to land reform (0 = reform year)

You have already seen an event study: Meng, Qian and Yared's Figure 3, around the famine

IS IT REALLY THE LAND REFORM?

- **The rival: the One Child Policy**, started at almost the same time. But not in the same year in each county: adding the county's One Child Policy date **does not change** the result, and the rise starts **before** the rural policy
- **Who has second children** did not change with the reform: the result is not a different mix of families
- **Why?** More consistent with **income**: larger effects among more educated parents and where output grew more
- **How?** Ultrasound existed in provincial capitals: larger rise in rural counties **connected by rail** to one; a rise also where none was reachable

ONE PAPER, THREE TOOLS

Tool	In Almond, Li and Zhang	What it keeps out
Compare like with like (matching)	births in the same county and the same year	anything that hit a county in a year: weather, a clinic, a campaign
Difference in differences, year by year	before and after each county's own reform	fixed differences between counties and families
Nature's lottery (instrument logic)	the sex of the first child	differences between families that want to select and families that do not

Each problem can be taken apart **separately**, inside the same paper

WHAT STARTED CHINA'S MIRACLE? THE VIEW FROM OUTER SPACE

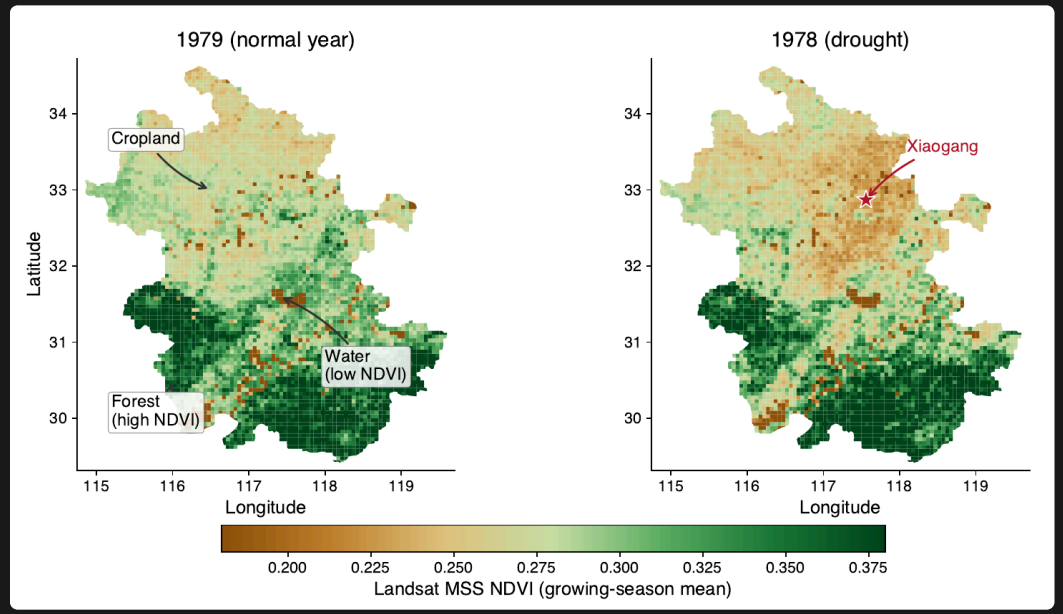
Joel Ferguson, Oliver Kim

Working paper, 2026

- **The standard story**: giving land back to households raised farm productivity and started the miracle. Lin (1992): almost **half** of farm output growth 1978-84
- It rests on **official statistics**, from a state whose local officials had over-reported harvests before, during the Great Leap
- And two problems we can now name: the early reformers were **poor and drought-hit**; prices **rose in 1979** for everybody
- **Question**: did reforming **sooner** raise grain **yields**?

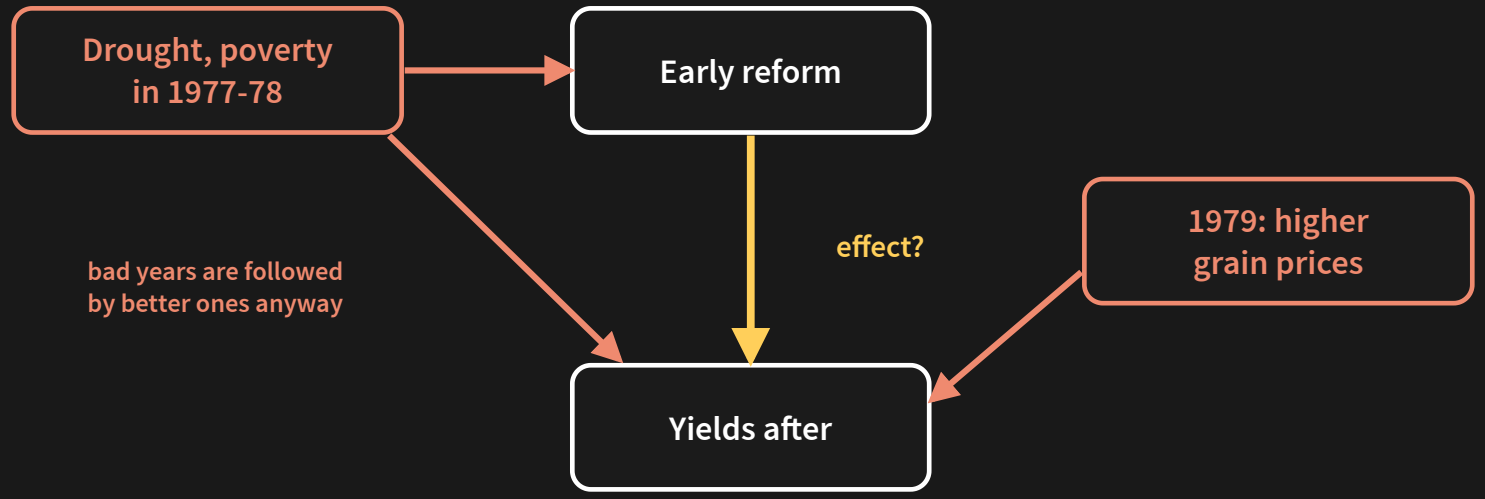
DATA FROM SPACE

- **Satellites**: Landsat (from 1972, dots of about 60 metres) and a coarser weather satellite
- **NDVI**: how green the fields are. Checked against crop yields in many countries and periods
- **Independent** of the Chinese statistical office
- Reform dates: official, by province (Lin 1992), and from county histories (Almond, Li and Zhang)



Ferguson and Kim, Fig. 2: Anhui seen by Landsat in 1979 (normal year) and 1978 (drought). The star: Xiaogang, where households started the reform

DID THE REFORM RAISE YIELDS? DRAW THE GRAPH FIRST

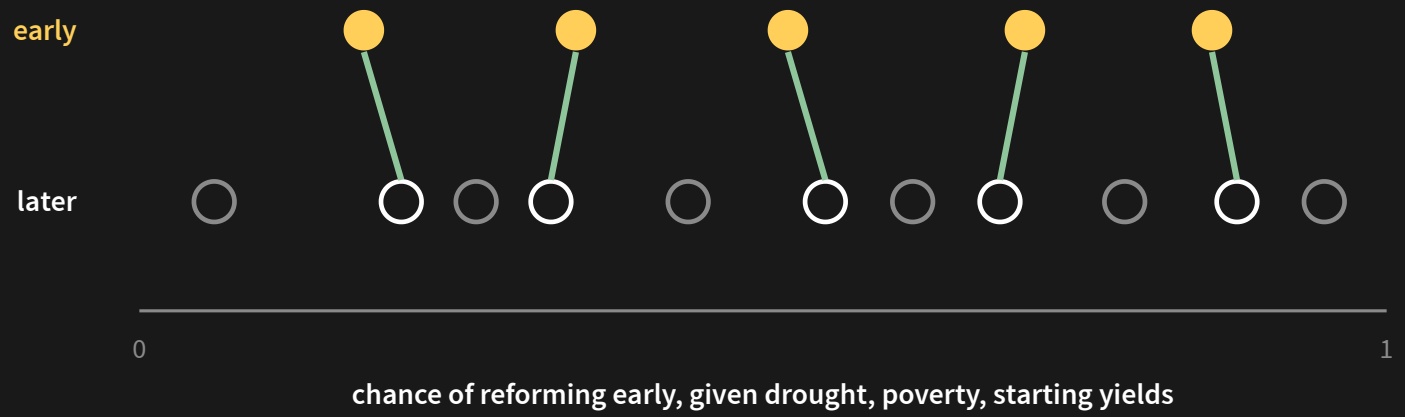


- The first to reform were the **poorest and drought-hit**: their yields would have bounced back anyway (**mean reversion**)
- And in **1979 prices rose for everybody**, at the same time as the reform spread

HOLDING MANY THINGS FIXED: REGRESSION, FOR DUMMIES

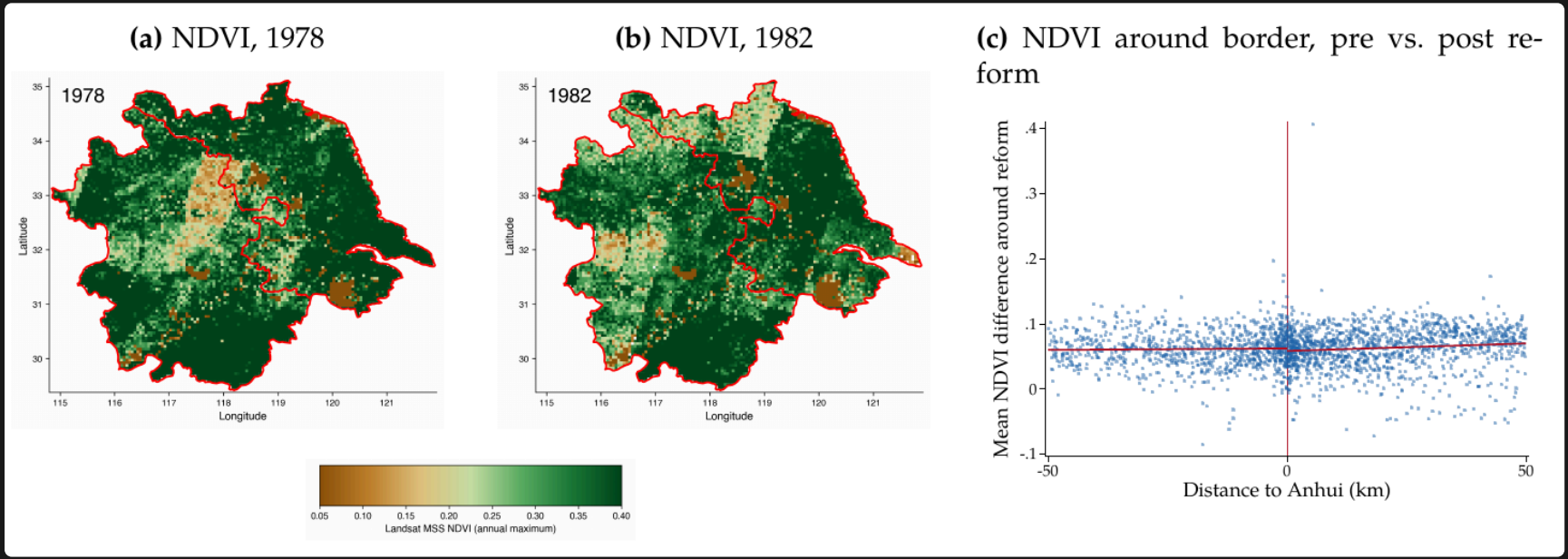
- Early and late reformers differ in **many** things we can measure: starting yields, drought, poverty, crops...
- Ideal: compare early and late reformers that are **identical** in all of them. With a handful of things, the groups empty fast (remember: 19 provinces, one per cell)
- **Regression** fills the empty cells by assuming a simple shape: each factor shifts yields by a fixed amount. It compares like with like, **on average**
- It only holds fixed **what we measure**. The luck of a good year, local politics, the 1979 prices: only if measured

PROPENSITY SCORE, FOR DUMMIES



- **Step 1**: from what we see, compute each county's **chance** of reforming early. Many things become **one number**
- **Step 2**: pair each early reformer with a later one that had a **similar chance**, and compare their yields
- Same assumption as matching: **nothing unseen** drives both timing and yields. Ferguson and Kim go further: **borders** and **event studies**

A BORDER SEEN FROM SPACE

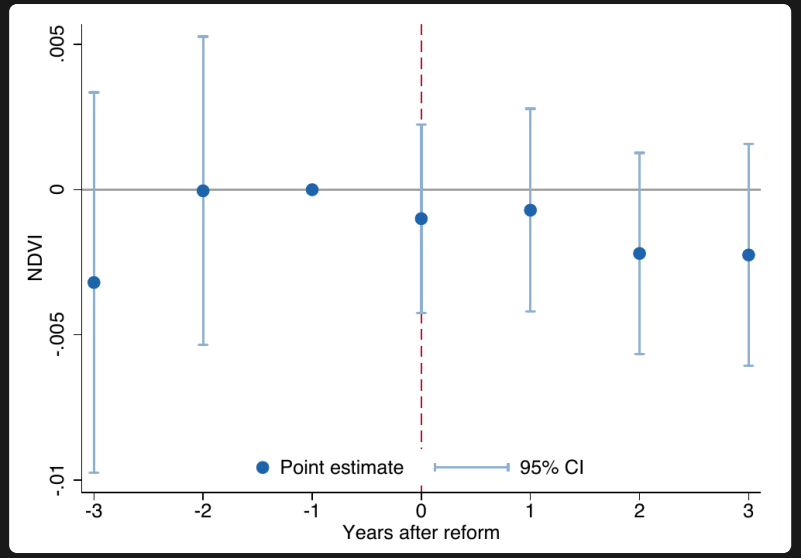


Ferguson and Kim (2026), Fig. 5: vegetation over Anhui and Jiangsu in 1978 and 1982; change in vegetation against distance to the Anhui border

- Anhui reformed first (1978), Jiangsu later. Compare fields **just across the border**, alike in soil and weather, **before and after** the reform: a difference in differences between neighbours
- If the reform raised yields, the Anhui side would have grown greener than the Jiangsu side: the right panel would show a **step at zero**. It is flat

OFFICIAL DATA AND THE VIEW FROM SPACE

- **Lin (1992)**, official provincial data: the reform explains **almost half** of farm output growth, 1978-84
- **Ferguson and Kim**: vegetation measured by **satellites** (Landsat, from 1972), not by the statistical office
- Same staggered timing, before and after: **no effect on yields**, even three years later
- What moved yields? Their evidence points to the **1979 price increase**; the reform instead freed labour for other work
- Data are political, again: the miracle's first chapter is being **rewritten with new data**



Ferguson and Kim (2026), Fig. 6: change in the border gap in vegetation (NDVI) after a province reforms, years 0-3

BACK TO THE TRANSITION

- We still cannot say what "the transition" caused. We can say something about its **pieces**:
- Sent-down years: **no lasting benefit** once siblings are compared
- Land reform: **more boys** born after a first girl; on yields, a **live debate** between official data and satellites, with prices as the rival
- Satellites can **overturn** a story built on official data
- Every method buys an answer with an **assumption**. The job is to say which, and to check it

QUESTIONS, FOR NEXT WEEK

Two papers, two briefs, one set of questions each:

- Almond, Li, Zhang (2019): land reform and the missing girls
- Chen, Fan, Gu, Zhou (2020): city youths in village schools

Read the brief (the paper if you like) and answer the questions: causal graphs, the ideal experiment, data, China, theory.

Please no AI chatbots: use the papers, the lectures and, only if needed, a search without AI (google.com/search?udm=14)



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Send your answers to francesco.manaresi@bancaditalia.it

They won't be graded!